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Declaration:

I hereby declare that the content of this assignment is original and has been prepared by me.

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Set -1

Question 1(a) :

Answer :

Statistics is the science of collecting, organizing, analyzing, interpreting, and presenting data. It helps convert raw data into meaningful insights and supports informed decision-making. Croxton and Cowden state, "Statistics is the science that deals with the collection, presentation, analysis, and interpretation of numerical data."

Scope in Various Fields:

- **Business:** Helps in sales forecasting and market analysis.
- **Healthcare:** Assists in clinical trials and patient data evaluation.
- **Education:** Used to analyze student performance and curriculum effectiveness.
- **Government:** Supports census, budgeting, and policy formulation.
- **Agriculture:** Applied in crop yield estimation and resource management.
- **Social Sciences:** Used for survey analysis and behavioral studies.
- **Engineering:** Helps in quality control and project planning.

(b)

Answer:

Data classification is the systematic arrangement of data into categories to simplify analysis. It enhances clarity, facilitates comparison, and aids in better data management.

Process:

1. Data Collection
2. Sorting and Grouping
1. Labeling Categories
2. Tabulating Data

Importance:

- Simplifies complex data
- Enhances comparison
- Improves clarity
- Supports effective decision-making

Attributes vs Variables:

- **Attributes:** Qualitative, e.g., gender, blood group
- **Variables:** Quantitative, e.g., age, height, income

(c)

Answer:

A frequency distribution organizes data into classes or intervals and shows how many values fall into each class. It is useful for summarizing and visualizing large datasets.

Advantages:

- Simplifies large datasets
- Highlights patterns and trends
- Supports graphical representation (histograms, bar charts)
- Aids in statistical analysis
- Facilitates comparison

Question 2 :**Answer:**

Central tendency refers to the measure that identifies the center of a dataset. It includes mean, median, and mode.

Purpose:

- Summarizes data into a single representative value
- Facilitates comparison
- Forms basis for further statistical analysis

Characteristics of a Good Measure:

- Clearly defined
- Simple to understand and calculate
- Considers all values
- Minimally affected by outliers
- Applicable to further analysis

(b)

Answer:**Median (Ungrouped):**

Data: 10, 20, 30, 40, 50

Median = Middle value = 30

Median (Grouped):

Use formula: $L + \left(\frac{\frac{N}{2} - F}{f} \right) \times h$

Mode (Ungrouped):

Data: 5, 6, 7, 5, 10

Mode = 5

Mode (Grouped):

Use formula: $L + \left(\frac{f_1 - f_2}{2f_1 - f_0 - f_2} \right) \times h$

(c)

Answer:

Arithmetic Mean: Simple average of all values. Weighted Mean: Each value is multiplied by a weight reflecting its importance.

Formula:

- Arithmetic Mean = $\frac{\sum x}{n}$
- Weighted Mean = $\frac{\sum wx}{w}$

Applications:

- Arithmetic Mean: Used in general analysis, e.g., average marks
- Weighted Mean: Used in index numbers, CGPA calculation

Question 3 (a):

Answer : Dispersion refers to the extent to which data values deviate from the central value. It shows the spread or variability in a dataset. The greater the dispersion, the more scattered the data is.

Significance:

- Helps understand the consistency of data
- Assists in risk and uncertainty analysis
- Essential for comparing datasets

Types of Measures:

- **Absolute Measures:** Represent the spread in original units (e.g., Range, Standard Deviation)
- **Relative Measures:** Express spread in percentage or ratio form, allowing comparison across datasets (e.g., Coefficient of Variation)

(b)

Answer:

Range: Range = Maximum Value - Minimum Value

Example: Data = 10, 15, 20 → Range = 20 - 10 = **10**

$$\text{Variance : } \text{Variance} = \frac{\sum(x-\bar{x})^2}{n}$$

Example: Data = 2, 4, 6

$$\text{Mean } \bar{x} = 4, \text{ Variance} = \frac{(2-4)^2 + (4-4)^2 + (6-4)^2}{3} = \frac{8}{3} = 2.67$$

Standard Deviation (SD): $\sqrt{\text{Variance}}$

Using the above example : SD : $\sqrt{2.67} \approx 1.63$

(c)

Answer:

Standard Deviation Scenario:

A teacher wants to evaluate which of two sections performed more consistently in an exam. Section A has SD = 2, and Section B has SD = 5. Lower SD in Section A indicates more consistent performance.

Relative Variance Scenario (Coefficient of Variation):

An investor compares two stocks: Stock X and Stock Y. Even if Stock Y has a higher return, a higher coefficient of variation (CV) indicates more risk. Hence, CV helps assess risk per unit return, aiding investment decisions.

Question 4(a) :

Answer : Probability is a branch of mathematics that quantifies uncertainty. There are three main approaches to understanding probability: classical, empirical, and subjective.

1. Classical Approach:

Also known as the theoretical approach, it is based on the assumption that all outcomes in a sample space are equally likely.

$$P(E) = \frac{\text{Number of favorable outcomes}}{\text{Total number of outcomes}}$$

Example: The probability of getting a 3 when rolling a fair six-sided die is .

3. Empirical Approach:

Also called the experimental approach, it is based on actual data collected through experiments or observations. It depends on the frequency of occurrence over repeated trials.

$$P(E) = \frac{\text{Numbers of times event occurred}}{\text{Total number of Trials}}$$

Example: If it rained 30 out of the last 100 days, the empirical probability of rain tomorrow is 0.30.

3. Subjective Approach:

This is based on personal judgment, intuition, or experience rather than numerical data or logical analysis.

Example: A doctor estimates a 70% chance of a patient's recovery based on past experience.

Significance in Analyzing Uncertainty:

- The classical method is useful in games of chance and mathematical problems.
- The empirical method is effective when historical data is available.
- The subjective method is applicable in decision-making where data is limited or unavailable.

Each method offers unique insight into uncertainty and is used depending on the context and data availability.

(b)

Answer : A sample space is the set of all possible outcomes of a random experiment. It is denoted by the symbol S .

Types of Sample Spaces:

1. **Finite Sample Space:** Contains a countable number of outcomes. Example: Tossing a coin once $\rightarrow S = \{\text{Heads, Tails}\}$
2. **Infinite Sample Space:** Contains an uncountable or endless number of outcomes. Example: Tossing a coin until heads appear $\rightarrow$ Infinite possible outcomes.
3. **Discrete Sample Space:** Outcomes are countable, even if large in number. Example: Number of students in a class $\rightarrow 1, 2, 3, \dots$
4. **Continuous Sample Space:** Outcomes can take any value within a range, including fractions or decimals. Example: Time taken to complete a task $\rightarrow 1.2 \text{ sec}, 1.35 \text{ sec}, \text{ etc.}$

Understanding the type of sample space helps in choosing the correct probability model.

(c)

Answer : In probability theory, an event is any specific outcome or a set of outcomes from a random experiment. An event is a subset of the sample space and is usually denoted by capital letters like A , B , or E .

Types of Events:

1. **Simple Event:** Involves only one outcome.
Example: Getting a 5 when rolling a die.
2. **Compound Event:** Involves two or more outcomes.
Example: Getting an odd number (1, 3, or 5) on a die.
3. **Sure Event:** The event that always happens.
Example: Getting a number less than 7 when rolling a die.
4. **Impossible Event:** The event that never happens.
Example: Getting a 7 when rolling a standard die.

Role of Events in Probability Theory:

- Events are essential to define and calculate probabilities.
- They help in understanding the likelihood of specific results in uncertain situations.
- Events allow statisticians and analysts to apply probability rules like the addition and multiplication principles.
- Events can be combined (unions or intersections) or complemented to analyze more complex problems.

Real-Life Example:

Scenario:

A student appears for a multiple-choice quiz with 4 questions. Each question has 4 options, only one of which is correct.

Event A: The student gets at least one question right.

Here, the sample space is all possible answer combinations, and event A represents a subset where at least one correct answer occurs. We can calculate the probability of A or even $P(A')$, where A' is the event of getting all answers wrong. Such an event helps students, teachers, or AI-based systems evaluate the chances of success in quizzes and tests.

Conclusion:

Events are the building blocks of probability theory. They provide the foundation for all probability calculations by defining the outcomes we are interested in. Whether simple or complex, understanding events helps in better decision-making in uncertain real-life scenarios like exams, games, medical treatments, and business risks.

Question 5 (a) :

Answer : In probability theory, an experiment refers to a process that leads to an outcome or a set of outcomes. Based on the predictability of results, experiments are classified into three types:

1. Deterministic Experiment:

- These are experiments whose outcomes are certain and predictable.
- Repeating the experiment under identical conditions always gives the same result.
- There is no element of chance or randomness.

Example:

- Boiling water at 100°C at sea level will always produce steam.
- Solving a math problem with a known formula gives the same answer every time.

Use in Probability:

- Not commonly used, as probability is concerned with uncertainty and variability.

2. Non-Deterministic (Random) Experiment:

- These experiments have uncertain outcomes.
- Even when repeated under the same conditions, the result may vary.
- These are the core of probability theory.

Example:

- Tossing a coin → Outcome: Head or Tail
- Rolling a die → Outcome: Any number from 1 to 6

Use in Probability:

- Essential in analyzing chance events, calculating probabilities, and modeling real-world situations like games, insurance, and risk assessment.

3. Hybrid Experiment:

- Combines both deterministic and random components.
- Part of the outcome is predictable, while another part is uncertain.

Example:

- A machine always starts when powered on (deterministic), but the number of defective items it produces during a shift may vary (random).

Use in Probability:

- Widely used in simulations, manufacturing, robotics, and quality control, where systems have predictable routines but unpredictable results.

Conclusion:

- Deterministic experiments are fixed and certain → No probability needed.
- Non-deterministic experiments are the heart of probability analysis.
- Hybrid experiments bridge the gap between the two and are important in real-life modeling.

(b)

Answer : The Expected Value (EV) is a core concept in probability that refers to the long-term average or predicted value of a random variable over many repetitions of an experiment. It is calculated as the weighted average of all possible outcomes, where each outcome is multiplied by its respective probability.

Formula:

$$EV = \sum(x_i \times P(x_i))$$

Where:

x_i = outcome

$P(x_i)$ = probability of the outcome

Example:

A game offers ₹100 for a win and ₹0 for a loss. The probability of winning is 0.3 and losing is 0.7.

$$EV = (100 \times 0.3) + (0 \times 0.7) = 30$$

So, the expected earning per game is ₹30.

Use in Decision-Making:

- Helps businesses evaluate investment opportunities
- Used in insurance to determine premiums
- Assists individuals in choosing between multiple risky options
- Allows for quantitative comparison of uncertain outcomes

Real-Life Example:

An investor chooses between two stocks:

- Stock A: 50% chance of gaining ₹10,000 or losing ₹5,000
- Stock B: 80% chance of gaining ₹5,000 or losing ₹1,000

Calculating EV for both helps decide which option has a better average return over time.

Conclusion:

Expected value gives a numerical basis for evaluating uncertain outcomes, making it a powerful tool for decision-making in business, finance, insurance, and everyday life.

(c)

Answer : In probability theory, understanding the nature of events is important for accurate calculations. Two such important concepts are equally likely events and exhaustive events.

Equally Likely Events:

Events are equally likely when each has the same chance of occurring.

Example:

In a fair die roll: Events {1, 2, 3, 4, 5, 6}

Each has a probability = $\frac{1}{6}$

Effect on Probability:

Probability of an event = $P(E) = \frac{\text{Number of favorable outcomes}}{\text{Total number of outcomes}}$

Equally likely assumption is essential for using the classical probability formula.

Exhaustive Events:

A set of events is exhaustive if it includes all possible outcomes of an experiment.

Example:

Tossing a coin $\rightarrow$ Events {Heads, Tails}

Together they exhaust all outcomes, so they are exhaustive.

Effect on Probability:

The total probability of exhaustive events must add up to **1**:

$$P(H) + P(T) = 1$$

Why Are These Important?

- Ensures the validity of probability calculations
- Helps define sample space clearly
- Avoids mistakes in counting outcomes

Combined Example:

In rolling a die:

- Events {1, 2, 3, 4, 5, 6} are exhaustive and equally likely.
- Probability of getting an even number =

$$P(\text{Even}) = \frac{3}{6} = 0.5$$

This calculation is possible only because the outcomes are both equally likely and exhaustive.

Conclusion:

Equally likely and exhaustive events form the foundation of classical probability. Accurate identification of these ensures correct probability results and better decisions in uncertain situations.

Question 6 (a) :

Answer : The addition rule of probability is used to find the probability of either of two events happening.

Addition Rule Formula:

1. For mutually exclusive events (events that cannot occur at the same time):

$$P(A \cup B) = P(A) + P(B)$$

2. For non-mutually exclusive events (events that can occur together):

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Mutually Exclusive Events:

- Events that cannot occur at the same time.
- Example: Drawing a red card and a black card in one draw $\rightarrow$ Not possible.

Example Calculation:

Tossing a coin →

$$P(\text{Head or Tail}) = P(\text{Head}) + P(\text{Tail}) = 0.5 + 0.5 = 1$$

Non-Mutually Exclusive Events:

- Events that can happen together.
- Example: Drawing a card that is a King or a Heart → Some cards are both.

Example Calculation:

$$P(\text{King or Heart}) = P(\text{Heart}) - P(\text{King and Heart})$$

$$= \frac{4}{52} + \frac{13}{52} - \frac{1}{52} = \frac{16}{52} = 0.31$$

Conclusion:

The addition rule helps calculate the combined probability of two events. It is important to first check if the events are mutually exclusive or not.

(b)

Answer:

The multiplication rule calculates the probability of two events occurring together (A and B).

Multiplication Rule Formula:**1. For Independent Events:**

$$P(A \cap B) = P(A) \times P(B)$$

Events that do not affect each other.

Example:

Tossing a coin and rolling a die:

$$P(\text{Head and 4}) = \frac{1}{2} \times \frac{1}{6} = \frac{1}{12}$$

2. For Dependent Events:

$$P(A \cap B) = P(A) \times P(B|A)$$

Events where one affects the other.

Example:

Picking 2 cards without replacement from a deck:

- $P(\text{First Ace}) = 4/52$
- $P(\text{Second Ace} | \text{First Ace}) = 3/51$
- $\text{Total} = \frac{4}{52} \times \frac{3}{51} = \frac{12}{2652} = 0.0045$

Conclusion:

Use the multiplication rule to find joint probabilities. Always check if events are independent or dependent before applying the formula.

(c)

Answer:

Conditional Probability is the probability of an event given that another event has already occurred.

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

It helps in analyzing situations where additional information is known.

Bayes' Theorem:

Bayes' Theorem updates probability based on new evidence. It links prior knowledge ($P(A)$) with new information ($P(B|A)$).

$$P(A|B) = \frac{P(B|A) \cdot P(A)}{P(B)}$$

Real-Life Applications:

1. **Medical Diagnosis:**

Bayes' Theorem is used to update the probability of a disease given a positive test result.

2. **Spam Filters:**

Email systems use it to detect spam based on past data.

3. **Machine Learning & AI:**

Bayesian algorithms help in predictions based on prior outcomes.

Conclusion:

Bayes' Theorem and conditional probability are essential tools for data-driven decision-making. They help professionals in healthcare, finance, and tech to make better predictions and smarter choices.